NATIONAL UNIVERSITY OF SINGAPORE

CS1231S – DISCRETE STRUCTURES

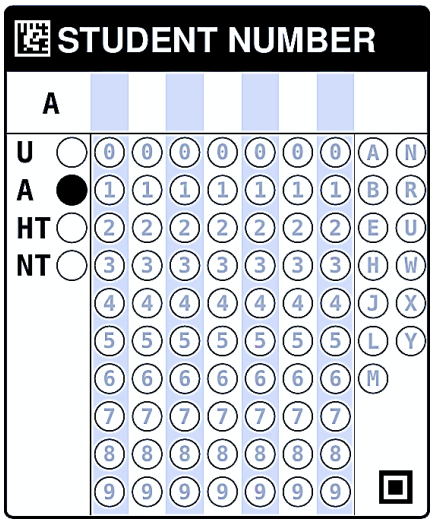
(Semester 1: AY2025/26)

Final Assessment Answer Sheets

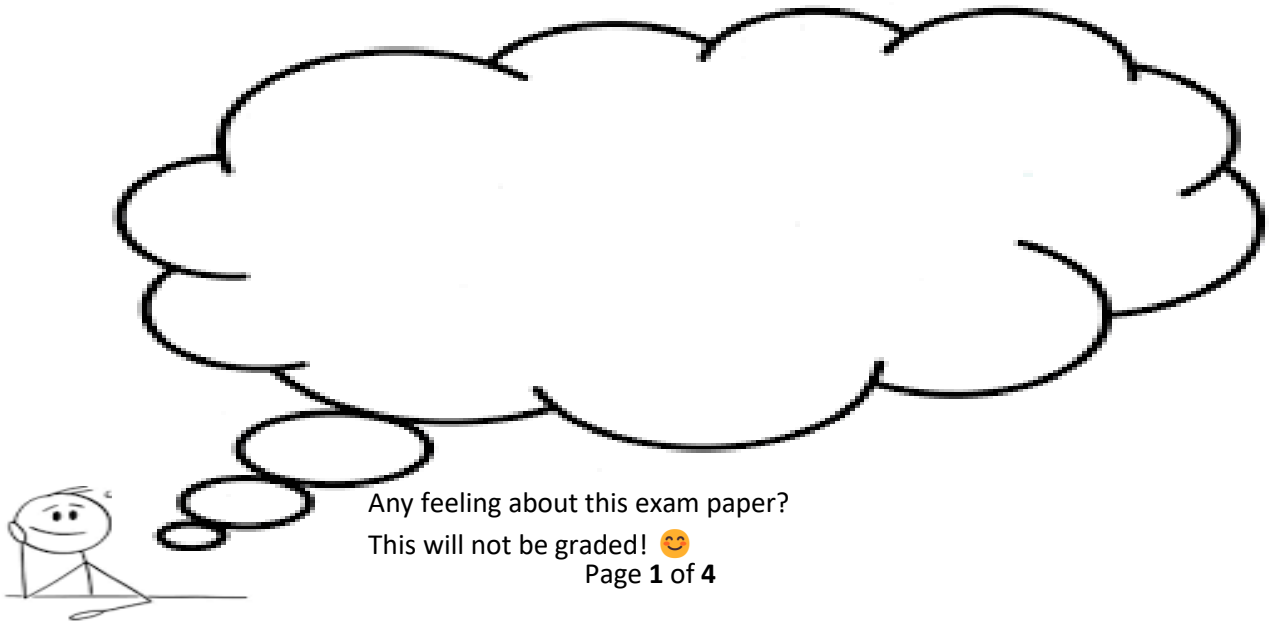
Time Allowed: 2 Hours

INSTRUCTIONS

1. Write your **Student Number** on the right AND, using a **pencil** (2B or above), shade the corresponding circle **completely** in the grid for each digit or letter. DO NOT WRITE YOUR NAME!
2. Zero mark will be given if you write/shade your Student Number incompletely or incorrectly.
3. Write your Student Number at the top of page 3.
4. There are **FOUR (4) pages** in the Answer Sheets.
5. All questions must be answered in the space provided; no extra sheets will be accepted as answers.
6. You must submit only these **ANSWER SHEETS** and no other documents.
7. An excerpt of the question may be provided to aid you in answering in the correct box. It is not the exact question. You should still refer to the original question in the question paper.
8. You may write your answers in pencil (2B or above) or pen. Pencil is preferred in case you need to erase and rewrite your answers.
9. The maximum mark for this paper is 100.
10. **Marks may be deducted** for (i) illegible handwriting, and/or (ii) excessively long answer.
11. Each multiple choice question is intended to have only one answer. Please shade the appropriate bubble in **pencil**.



For Examiner's Use Only		
Question	**Marks**	**Remarks**
Q1-23	/ 46	
Q24	/ 20	
Q25	/ 20	
Q26	/ 14	
Total	**/ 100**	



Any feeling about this exam paper?
This will not be graded! 😊

Part A: Multiple Choice Questions (Total: 46 marks)

Please shade only ONE bubble for each question using **pencil (2B or above)**.

	(A)	(B)	(C)	(D)	(E)	
1.	○	○	○	○	○	1.
2.	○	○	○	○	○	2.
3.	○	○	○	○	○	3.
4.	○	○	○	○	○	4.
5.	○	○	○	○	○	5.
6.	○	○	○	○	○	6.
7.	○	○	○	○	○	7.
8.	○	○	○	○	○	8.
9.	○	○	○	○	○	9.
10.	○	○	○	○	○	10.
11.	○	○	○	○	○	11.
12.	○	○	○	○	○	12.
13.	○	○	○	○	○	13.
14.	○	○	○	○	○	14.
15.	○	○	○	○	○	15.
16.	○	○	○	○	○	16.
17.	○	○	○	○	○	17.
18.	○	○	○	○	○	18.
19.	○	○	○	○	○	19.
20.	○	○	○	○	○	20.
21.	○	○	○	○	○	21.
22.	○	○	○	○	○	22.
23.	○	○	○	○	○	23.
	(A)	(B)	(C)	(D)	(E)	

Student Number: A | | | | | | | | |

Part B (Total: 54 marks)

24. [20]

(a) 12 balls [2]

(b) Committees [2]

(c) Passwords [2]

(d) Circular table [2]

(e) Circular table [2]

(f) Socks [2]

(g) 3 dice [3]

(h) (i) $P(S|O)$ [2] (ii) $P(S|U)$ [2] (iii) Spam signal [1]

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25. [20]

(a) Is $K_{3,4}$ planar? [2]

(b) $K_{m,n}$ to be Eulerian [2]

(c) $K_{m,n}$ to be Hamiltonian [2]

(d) Which cycle graphs are bipartite graphs [2]

(e) Perfect matching condition [2]

(f) Draw bipartitite graph and write out the maximal matching and maximum matching [4]

(g) Is there a perfect matching for G_1? [3]

(h) Is there a perfect matching for G_2? [3]

26.

[14]

(a)

(i) f [2]

(ii) g [2]

(iii) h [2]

(iv) k [2]

(b)

(i) $z^{-1}([3])$ [2]

(ii) $z^{(2)}(x)$ [2]

(iii) Order of [3] [2]

=== END OF PAPER ===